

Normal operator

Lemma. Let  $T: V \rightarrow V$  be a linear operator on fin. dim inner product space  $V$ .

If  $\lambda$  is an eigenvalue of  $T$ , then  $\overline{\lambda}$  is an eigenvalue of  $T^*$ .

Proof. Let  $v$  be an eigenvector of  $\lambda$ . Then, for any  $x \in V$ ,

$$0 = \langle 0, x \rangle = \langle (T - \lambda I)(v), x \rangle = \langle v, (T - \lambda I)^*(x) \rangle$$

That is,  $v \in (\text{Im}(T^* - \overline{\lambda}I))^{\perp} = \ker(T^* - \overline{\lambda}I) \neq \{0\}$ .

Schur's Lemma

Let  $T: V \rightarrow V$  be a linear operator on fin. dim inner product space  $V$  over a field  $F$ .

If a characteristic polynomial  $f(t)$  of  $T$  splits over  $F$ ,

then there exists an orthonormal basis  $\gamma$  for  $V$  s.t.  $[T]_{\gamma}$  is upper-triangular matrix.

Def. Let  $T: V \rightarrow V$  be a linear operator on inner product space  $V$ .

If  $T^* \circ T = T \circ T^*$ , then  $T$  is called Normal operator

Proposition, Let  $T: V \rightarrow V$  be a Normal operator on inner product space  $V$ .

1) For any  $x \in V$ ,  $\|T(x)\| = \|T^*(x)\|$ .

2) For any  $c \in \mathbb{F}$ ,  $T - cI$  is Normal.

3) For any  $\lambda \in \mathbb{F}$  and  $x \in V$ ,  $T(x) = \lambda x \Leftrightarrow T^*(x) = \bar{\lambda}x$

4) For any distinct eigenvalue  $\lambda_1, \lambda_2$  of  $T$ ,  $x \in E_{\lambda_1}$  and  $y \in E_{\lambda_2} \Rightarrow \langle x, y \rangle = 0$ .

Hermitian

Let  $V$  be an Inner product space.

Prop. Let  $T, U: V \rightarrow V$  be Hermitian.

$T \circ U$  is Hermitian iff  $T \circ U = U \circ T$

Proof.  $T \circ U$  is Hermitian  $\Leftrightarrow T \circ U = (T \circ U)^* = U^* \circ T^* = U \circ T$ .

Prop. Let  $T: V \rightarrow V$  be a linear operator, and  $W \subseteq V$  be a  $T$ -invariant subspace.

1. If  $T$  is Hermitian, then a restriction  $T|_W$  is Hermitian.

2.  $W^\perp$  is  $T^*$ -invariant subspace

3. If  $W$  is  $T, T^*$ -invariant,  $T|_W^* = T^*|_W$ .

4. If  $W$  is  $T, T^*$ -invariant and  $T$  is Normal, then  $T|_W$  is Normal.

Proof. 1) is clear: for any  $x, y \in W$ ,

$$\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T^*(y) \rangle = \langle x, T(y) \rangle = \langle x, T|_W(y) \rangle$$

2). Let  $x \in W^\perp$  be given. Then, for  $y \in W$ ,

$$\langle T^*(x), y \rangle = \langle x, T(y) \rangle = 0, \text{ because } T(y) \in W. \text{ Thus } T^*(x) \in W^\perp.$$

3). Let  $x, y \in W$  be given. Then,

$$\begin{aligned} \langle T(x), y \rangle &\stackrel{1)}{=} \langle T|_W(x), y \rangle = \langle x, T|_W^*(y) \rangle \\ &\stackrel{2)}{=} \langle x, T^*(y) \rangle = \langle x, T^*|_W(y) \rangle. \end{aligned}$$

Thus is, for any  $x, y \in W$ ,

$$\langle x, T|_W^*(y) - T^*|_W(y) \rangle = 0.$$

This implies  $T|_W^* = T^*|_W$

4) Directly, given by 3.

Prop. Let  $V$  be a fin. dim. Complex inn. prod. space, let  $T: V \rightarrow V$  be a Normal.

If  $W \subseteq V$  is  $T$ -invariant, then also  $T^*$ -invariant.

Proof. Since  $T$  is Normal operator on fin. dim. Comp. inn space,  $T$  is diagonalizable by an orthonormal  $\beta$ .

Let  $x \in W$  be given with a representation  $\sum a_i v_i$  where  $v_i \in \beta$ . Then

$$T^*(x) = \sum a_i T^*(v_i) = \sum a_i \bar{\lambda}_i v_i \in W, \text{ since } v_i \in W.$$

Prop. If  $T: V \rightarrow V$  is Normal, then

$$T(x) = 0 \Leftrightarrow \|T(x)\| = 0 \Leftrightarrow \|T^*(x)\| = 0 \Leftrightarrow T^*(x) = 0.$$

Thus,  $\ker T = \ker T^*$ .

Moreover, if  $V$  is fin. dim, then

$$\operatorname{Im} T = \operatorname{Im} T^*$$

Thm. Let  $V$  be a fin. dim, inn. space,  $T: V \rightarrow V$  be a Hermitian.

For any  $x \in V$ ,

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2.$$

Therefore,  $T - iI$  is Invertible and  $((T - iI)^{-1})^* = (T + iI)^{-1}$ .

Proof. By Lemma,

$$\|T(x) \pm ix\|^2 = \|T(x)\|^2 + \|x\|^2 \pm 2\operatorname{Re}\langle T(x), ix \rangle$$

Meanwhile, for any  $x \in V$ ,

$$\langle T(x), ix \rangle = -i\langle T(x), x \rangle$$

$$\langle T^*(x), ix \rangle = \langle x, iT(x) \rangle = -i\langle x, T(x) \rangle$$

This implies  $\operatorname{Re}\langle T(x), ix \rangle = 0$ .

Now,

$$\begin{aligned} \ker(T \pm iI) &= \{x \in V \mid (T \pm iI)(x) = 0\} \\ &= \{x \in V \mid \|T(x)\|^2 + \|x\|^2 = 0\} \\ &= \{0\}. \end{aligned}$$

Prop. Let  $V$  be a Complex inner prod space.

1)  $T$  is Hermitian  $\Leftrightarrow$  for all  $x \in V$ ,  $\langle T(x), x \rangle \in \mathbb{R}$ .

2) If for all  $x \in V$ ,  $\langle T(x), x \rangle = 0$ , then  $T = T_0$ .

Proof. At first, show that 2). For any  $x, y \in V$ ,

$$0 = \langle T(x+y), x+y \rangle = \langle T(x), x \rangle + \langle T(x), y \rangle + \langle T(y), x \rangle + \langle T(y), y \rangle = \langle T(x), y \rangle + \langle T(y), x \rangle$$

$$0 = \langle T(x+iy), x+iy \rangle = \langle T(x), x \rangle + \langle T(x), iy \rangle + \langle T(iy), x \rangle + \langle T(iy), iy \rangle = \langle T(x), iy \rangle + \langle T(iy), x \rangle$$

$$\text{Now, } \langle T(x), y \rangle + \langle T(y), x \rangle = 0 \text{ and } -\langle T(x), y \rangle + \langle T(y), x \rangle = 0.$$

That is,  $\langle T(x), y \rangle = 0$ . Since  $x, y$  are chosen arbitrarily,  $T = T_0$ .

Suppose  $T$  is Hermitian. Then, for any  $x \in V$ ,

$$\langle T(x), x \rangle = \langle x, T^*(x) \rangle = \langle x, T(x) \rangle = \overline{\langle T(x), x \rangle}$$

Thus  $\langle T(x), x \rangle \in \mathbb{R}$ .

Conversely, Suppose for all  $x \in V$ ,  $\langle T(x), x \rangle \in \mathbb{R}$ . Then, directly  $\langle T^*(x), x \rangle \in \mathbb{R}$ .

$$\begin{aligned} \text{Now } \langle (T - T^*)(x), x \rangle &= \langle T(x), x \rangle - \langle T^*(x), x \rangle \\ &= \langle T(x), x \rangle - \langle x, T(x) \rangle \\ &= 0. \end{aligned}$$

By 2), we obtain the result.

Prop.  $A = B^t B$  for some  $B \in M_n(\mathbb{R})$ ,  $A$  is called Gramian matrix.

$A$  is Gramian if and only if  $A^t = A$  and  $A$  has no negative eigenvalue.

Proof. Suppose that  $A$  is Gramian. Then, clearly symmetric.

$$\text{Since } L_A = L_{B^t B} = (L_{B^t}) \circ (L_B),$$

$$\langle L_A(x), x \rangle = \langle L_{B^t B}(x), x \rangle \geq 0.$$

Conversely, For each  $1 \leq i, j \leq n$ ,

$$\left( [U^*]_{\beta/\alpha} \right)_{ij} = \langle U^*(v_i), v_j \rangle = \langle v_j, U(v_i) \rangle = \sqrt{\lambda_i} \cdot \langle v_j, v_i \rangle = \sqrt{\lambda_i} \delta_{ij}$$

$$\text{Meanwhile, } T(v_i) = (U^* \circ U)(v_i) \quad [T]_{\beta} = A = [U]_{\beta}^* [U]_{\beta}.$$

Def. Let  $V$  be a fin. dim. inn. space,  $T: V \rightarrow V$  be a Hermitian.  $T$  is called positive semidefinite if

$$\forall x \in V, \langle T(x), x \rangle \geq 0$$

In addition, if  $\langle T(x), x \rangle = 0 \Leftrightarrow x = 0$ , then  $T$  is called positive definite.

Thm. Let  $V$  be an inn. space with  $\dim V = n$ , and  $\beta$  be an orthonormal basis for  $V$ . Let  $T: V \rightarrow V$  be Hermitian,  $A = [T]_{\beta}$ .

1).  $T$  is positive semidefinite if and only if every eigenvalue of  $T$  is non-negative.

2)  $T$  is positive definite if and only if every eigenvalue of  $T$  is positive.

Proof. Let  $\gamma$  be an orthonormal basis consided by eigenvectors of  $T$ .

Let  $x \in V$  be given. Perote  $\gamma = \{v_1, \dots, v_n\}$ ,  $B = [T]_{\gamma}$ . Then,

$$x = \sum_{i=1}^n a_i v_i \quad \text{for some } a_i \in F.$$

$$\begin{aligned} \text{Now, } \langle T(x), x \rangle &= \langle T(\sum_{i=1}^n a_i v_i), \sum_{i=1}^n a_i v_i \rangle \\ &= \langle \sum_{i=1}^n a_i \lambda_i v_i, \sum_{i=1}^n a_i v_i \rangle \\ &= \lambda_i \cdot \langle \sum_{i=1}^n a_i v_i, \sum_{i=1}^n a_i v_i \rangle \\ &= \sum_{i=1}^n \lambda_i \cdot \|a_i v_i\|^2 \end{aligned}$$

3).  $T$  is positive definite if and only if for any  $(a_1, \dots, a_n) \in F^n$ ,  $\sum_{i=1}^n A_{ii} a_i \bar{a}_i > 0$ .

Proof. Let  $\beta = \{w_1, \dots, w_n\}$ . Then, for  $\underline{\sum a_i w_i}$ ,

$$\begin{aligned} &\langle T(\sum a_i w_i), \sum a_j w_j \rangle \\ &= \sum_{i,j} a_i \bar{a}_j \langle T(w_i), w_j \rangle \\ &= \sum \underline{A_{ji}} a_i \bar{a}_j \end{aligned}$$

Prop. If  $T^2 = U^2$  and  $T, U$  are positive semidefinite, then  $T = U$ .

Proof. Let  $\beta = \{v_1, \dots, v_n\}$  be an orthonormal consisted eigen -  $T$ .

For each  $1 \leq i \leq n$ ,  $T^2(v_i) = \lambda_i^2 v_i = U^2(v_i)$ .

If  $\lambda_i = 0$ , then clearly  $T(v_i) = 0$ , and

$$0 = \langle U^2(v_i), v_i \rangle = \langle U(v_i), U(v_i) \rangle$$

Thus  $U(v_i) = 0$ .

Now, assume  $\lambda_i > 0$ .

$$0 = (U^2 - \lambda_i^2 I)(v_i) = (U + \lambda_i I)(U - \lambda_i I)(v_i)$$

Since  $U$  is positive-semidefinite, every eigenvalue is bigger than zero.

Thus  $U + \lambda_i I$  is invertible. Now,  $(U - \lambda_i I)(v_i) = 0$ .  $\therefore U(v_i) = \lambda_i v_i = T(v_i)$ .